

ROBOCODE

PROJECT ABSTRACT: QUANT v6.0 LINE FOLLOWER

Team: ROBOCODE (Varna, Bulgaria)

Category: Line Follower Robot

Institution: ROBOCODE Robotics And Programming School (Varna)

Website: robocode.bg

Project Lead: Ruslan Kocherga, Director of ROBOCODE Varna

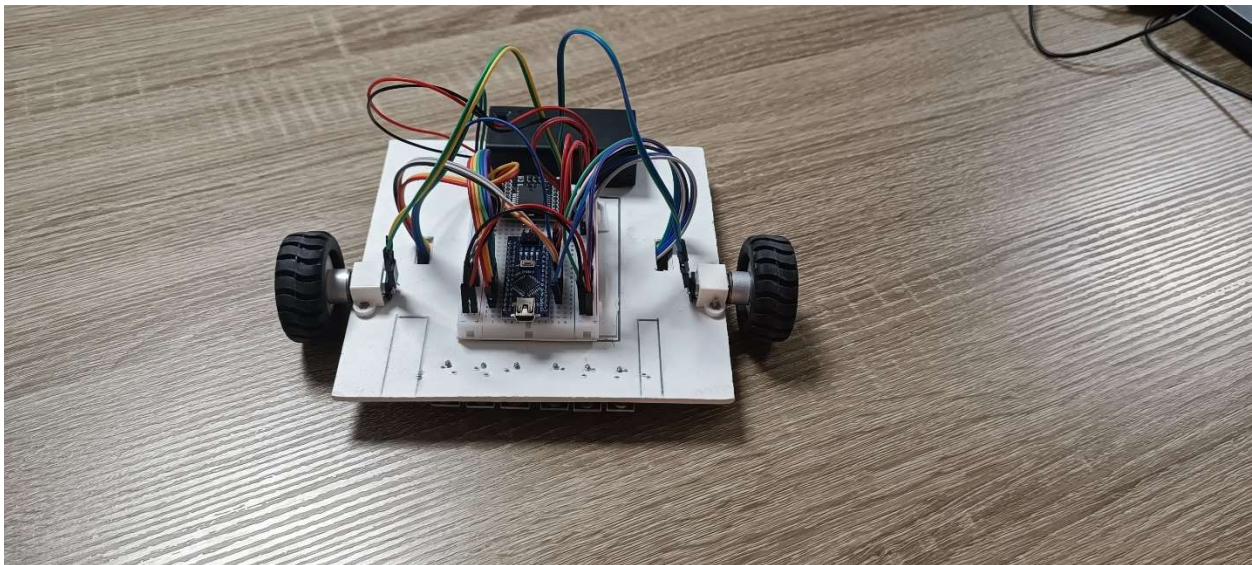
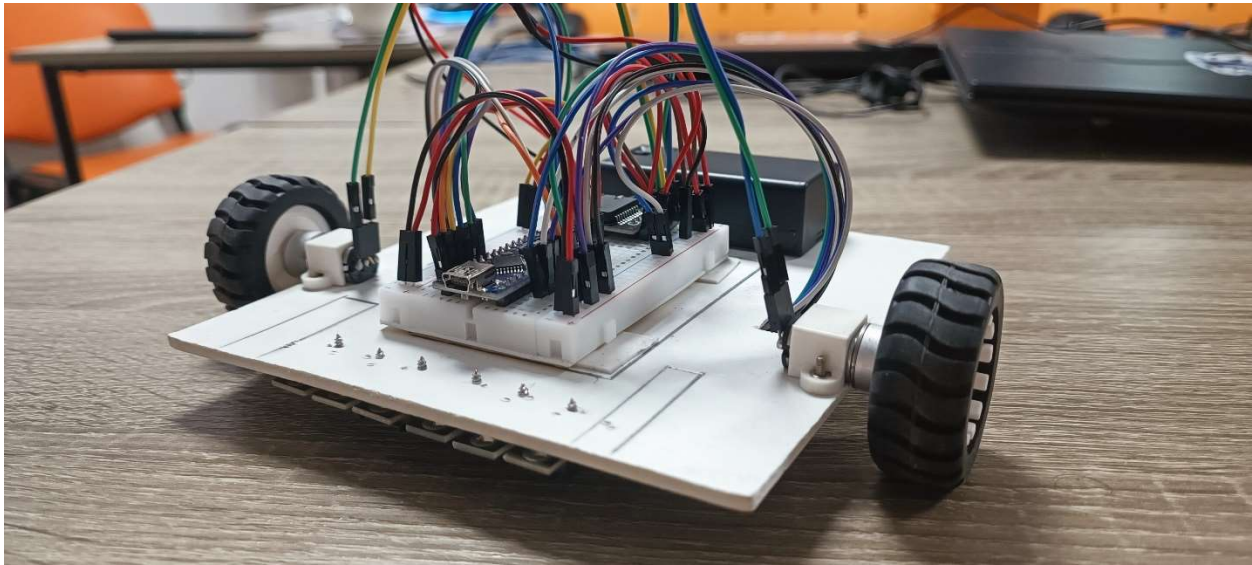
1. Project Overview

This project presents **Quant v6.0**, a high-performance autonomous racing platform developed at the **ROBOCODE** Robotics and Programming School in Varna, Bulgaria. The robot represents the culmination of advanced engineering and algorithmic optimization within the school's ecosystem.

While the standard robotics curriculum at **ROBOCODE** for 3rd-year students introduces a foundational 2-sensor robot designed for basic navigation (such as 90-degree turns and simple zig-zags), the **v6.0 "Competition Edition"** is a complete technological leap. It was re-engineered and programmed from the ground up by the team and students, under the technical mentorship of the school's director, **Ruslan Kocherga**.

The transition to version 6.0 involved replacing basic digital logic with a sophisticated analog sensing array and a custom **PD (Proportional-Derivative)** control system. This evolution was driven by the goal of competing on the international stage at **INFOMATRIX**, moving beyond classroom exercises toward professional-grade robotics. **Quant v6.0** is not just a racer; it is a demonstration of how modular hardware, when paired with advanced mathematical tuning, can achieve the precision and speed required for world-class line-following competitions.

1: General front/side view of Quant v6.0

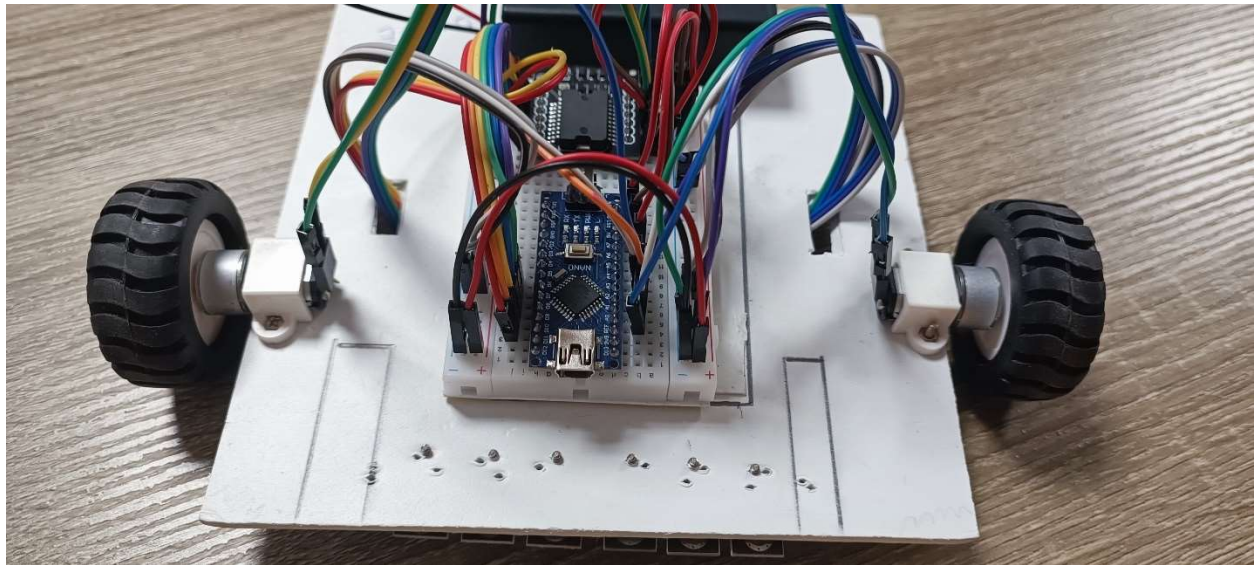


2. Evolution and Hardware Architecture

The transition to v6.0 required significant hardware upgrades to handle competitive speeds. The core architecture includes:

- **Microcontroller:** Arduino Nano (ATmega328P).
- **Motor Control:** A powerful L298P H-Bridge Driver, managing two high-RPM DC motors.
- **Sensor Array:** A front-mounted array of **6 Infrared Sensors (KY-033)**. Unlike basic educational robots using digital signals, Quant v6.0 connects these 6 sensors to **6 Analog Inputs**, allowing for high-resolution reflectivity data and precise calculation of the robot's position relative to the line's edge.

3: Bottom or detailed view showing the 6-sensor array and wiring



4. Software and PD Algorithm Logic

The software for **Quant v6.0** was developed by Mihail Vasilev and the team under the technical partnership and guidance of the project lead, **Ruslan Kocherga**. It implements a sophisticated **Proportional-Derivative (PD) Control Loop**, allowing the robot to process data and make hundreds of micro-corrections per second.

- **Weighted Error Calculation:** Each of the 6 sensors is assigned a mathematical weight. This creates a "weighted error" value, allowing the robot to accurately measure its distance from the line's center.
- **Proportional (Kp) - The "Correction" Term:** This coefficient handles immediate steering. If a sharp turn is detected (e.g., by sensors A0 or A5), Kp generates a powerful steering command to correct the course.

- **Derivative (Kd) - The "Damping" Term:** This coefficient analyzes how fast the robot is moving towards or away from the line. It acts as a "counter-force" to the Proportional term, preventing overshooting and eliminating "fishtailing" on straight sections, allowing for stable high-speed performance.

5. Technical Specifications

- **Input Sensing:** 6-Channel Analog IR Sensing (A0-A5).
- **Control Algorithm:** PD Feedback Loop (approx. 1000 Hz refresh rate).
- **Steering Logic:** Active Differential Steering with power compensation.
- **Safety Features:** Automated "Line Lost" recovery logic.
- **Platform:** ROBOCODE "Quant" proprietary chassis v6.0.

5. Conclusion

The "**Quant v6.0**" project demonstrates the successful escalation of a foundational educational platform into a high-performance racing robot. By combining a 6-sensor analog array with a finely tuned PD algorithm, the **ROBOCODE** Varna team has created a stable yet aggressive competitor, showcasing advanced engineering skills learned at our robotics school.